Flight Simulation of the AEGIS-75 High-Power Rocket Using RocketPy

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Abstract

This project presents the design and numerical simulation of the AEGIS-75 high-power sounding rocket using the RocketPy simulation framework. The objective was to evaluate the rocket's aerodynamic stability, flight trajectory, propulsion performance, and recovery sequence under realistic atmospheric conditions. A custom rocket model was developed using an AeroTech M1850W solid rocket motor, dual-deployment parachute recovery, and custom aerodynamic drag curves. The simulation predicts an apogee of approximately 700 m with stable flight throughout powered ascent and recovery.

1. Introduction

High-power rockets are widely used for educational research, atmospheric studies, and experimental payload testing. Before physical fabrication and launch, computer-based simulations allow engineers to evaluate vehicle performance, verify stability, and estimate flight characteristics under realistic environmental conditions.

RocketPy is an open-source six-degree-of-freedom flight simulator capable of modelling propulsion, aerodynamics, atmospheric conditions, and recovery systems. In this project, RocketPy was used to simulate the complete flight of the AEGIS-75 rocket. **Figure 1** shows the geometric configuration of the AEGIS-75 rocket.

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2. Objectives

The objectives of this project are:

- Design a 75 mm diameter high-power rocket.
- Model the rocket using RocketPy.
- Simulate the complete flight profile.
- Evaluate propulsion and aerodynamic performance.
- Determine maximum altitude, velocity, acceleration, and stability.
- Demonstrate parachute recovery.

3. Rocket Specifications

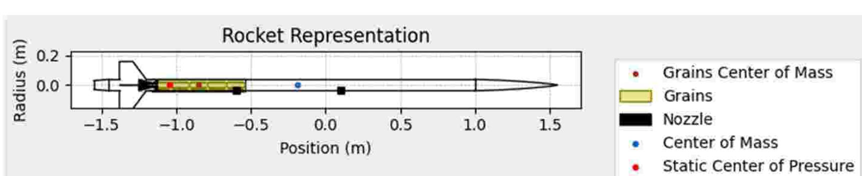

Figure 1. AEGIS-75 rocket geometry showing the nose cone, fin configuration, motor position, and recovery system

Parameter	Value
Rocket Name	AEGIS-75
Diameter	75 mm
Length	3.10 m
Empty Mass	18.0 kg
Loaded Mass	22.866 kg
Nose Cone	Von Kármán
Fin Configuration	Four Trapezoidal Fins
Recovery System	Dual Deployment
Launch Rail Length	6.0 m
Launch Angle	85°

4. Motor Specifications

The rocket uses an AeroTech M1850W solid rocket motor.

Parameter	Value
Motor Type	Solid Rocket Motor
Burn Time	4.01 s
Propellant Mass	2.956 kg
Average Thrust	1679.50 N
Maximum Thrust	2411 N
Total Impulse	6734.78 Ns
Average Exhaust Velocity	2278.41 m/s

5. Launch Environment

The launch site was modelled using forecast atmospheric data.

Parameter	Value
Latitude	19.15025°
Longitude	73.23245°

Parameter	Value
Elevation	72.6 m
Atmospheric Model	Forecast (GFS)
Surface Temperature	298.13 K
Surface Wind Speed	2.48 m/s

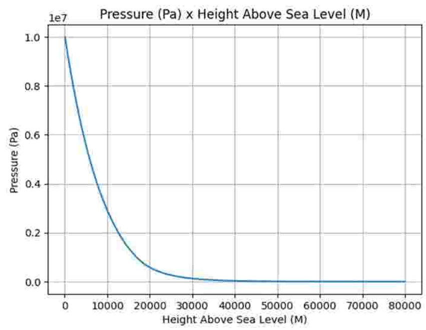

Figure 2. Pressure vs Altitude Plot

6. Aerodynamic Configuration

The rocket was equipped with:

- Von Kármán nose cone
- Four trapezoidal fins
- Boat tail
- Custom power-on and power-off drag curves
- Dual deployment parachute recovery

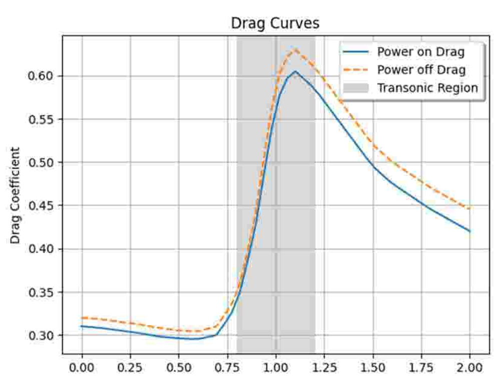

Figure 3. Aerodynamic drag coefficient variation with Mach number

The calculated stability margins were:

Condition	Static Margin
Launch	11.389 calibers
Rail Exit	11.514 calibers
Burnout	12.753 calibers

The positive stability margins indicate that the rocket remains statically stable throughout powered flight.

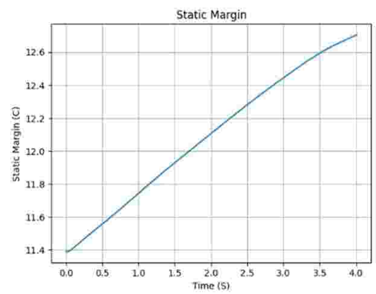

Figure 4. Static stability margin throughout the flight

7. Flight Performance

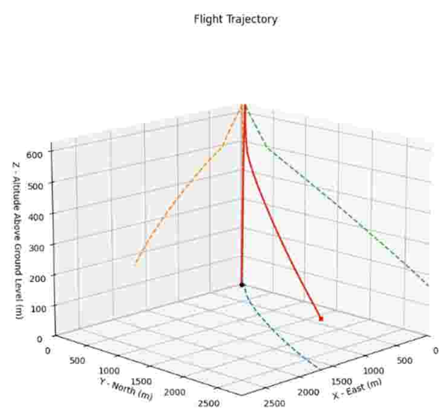

Figure 5. Three-dimensional simulated flight trajectory of the AEGIS-75 rocket

Launch

The rocket departed the launch rail after 0.377 s at a velocity of 27.36 m/s.

Powered Flight

During the powered ascent:

- Burn time: 4.01 s
- Burnout altitude: 385.41 m AGL
- Burnout speed: 118.58 m/s
- Burnout Mach number: 0.344

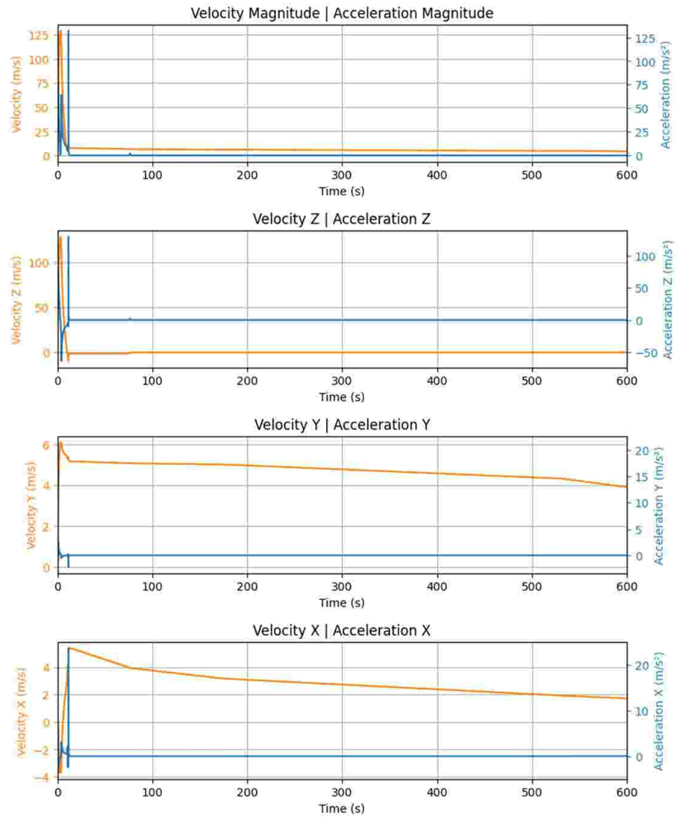

Figure 6. Linear kinematic response of the rocket during ascent and recovery

Coast Phase

After motor burnout, the rocket continued to climb due to its momentum before reaching apogee.

Apogee was reached after 10.035 s.

Maximum altitude:

699.83 m (ASL)

or

627.27 m (AGL)

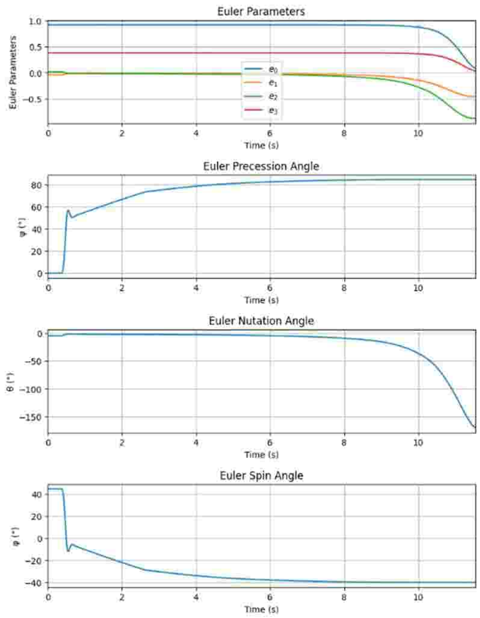

Figure 7. Euler parameters throughout the flight

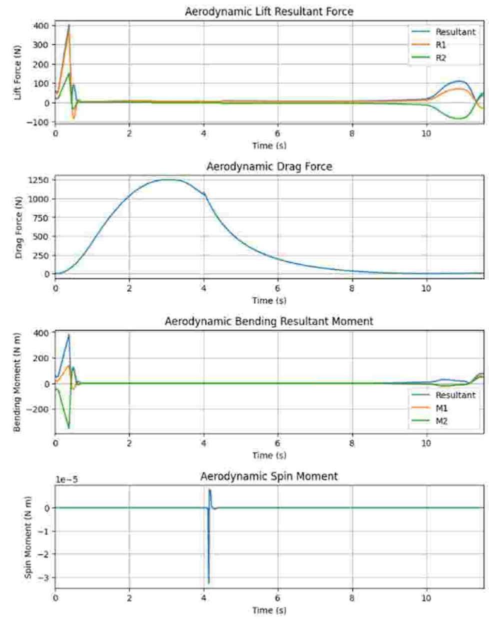

Figure 8. Aerodynamic forces acting on the rocket during flight

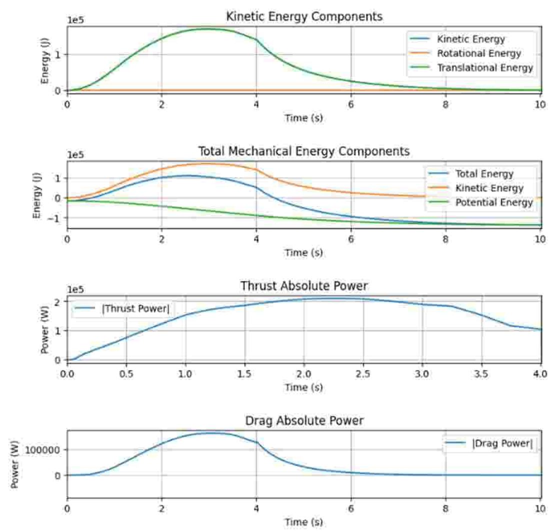

Figure 9. Variation of kinetic and potential energy during the flight

8. Recovery System

A dual-deployment recovery system was implemented.

Drogue Parachute

- Trigger: Apogee
- Inflation Time: 11.54 s

Main Parachute

- Trigger Altitude: 500 m AGL
- Inflation Time: 76.28 s

Both parachutes deployed successfully during the simulation.

9. Performance Summary

Parameter	Result
Maximum Altitude	699.83 m
Maximum Speed	128.90 m/s
Maximum Mach Number	0.373
Maximum Dynamic Pressure	9.45×10^5 Pa
Maximum Motor Acceleration	95.79 m/s^2
Maximum Motor Acceleration	9.77 g
Rail Exit Velocity	27.36 m/s
Burn Time	4.01 s

10. Stability Analysis

The simulation indicates that the centre of pressure remained behind the centre of gravity during the entire powered flight.

Static margin increased from:

11.389 calibers

to

12.753 calibers

showing excellent longitudinal stability.

11. Discussion

The RocketPy simulation successfully predicted the complete flight profile of the AEGIS-75 rocket.

The rocket remained aerodynamically stable throughout powered ascent, coast, and recovery. The predicted maximum velocity remained below the speed of sound (Mach 0.373), indicating a completely subsonic flight regime.

The dual-deployment recovery system operated correctly according to the programmed trigger conditions.

The simulation demonstrates the usefulness of RocketPy as a preliminary design and analysis tool for high-power rockets.

12. Conclusion

A complete numerical simulation of the AEGIS-75 high-power rocket was successfully carried out using RocketPy.

The rocket demonstrated stable flight characteristics, safe rail departure, successful motor operation, and reliable parachute deployment. The predicted apogee of approximately 700 m and the stable aerodynamic behaviour indicate that the proposed design is suitable as a preliminary high-power rocket configuration.

Future work may include experimentally validated drag coefficients, refined mass properties, CFD-based aerodynamic analysis, and comparison with OpenRocket or flight-test data to improve prediction accuracy.

References

1. RocketPy Documentation
2. AeroTech Reloadable Motor Data
3. Sutton, G. P., *Rocket Propulsion Elements*
4. Barrowman, J. S., *The Practical Calculation of the Aerodynamic Characteristics of Slender Finned Vehicles*